

Antifouling Properties of Poly(methyl methacrylate) Films Grafted with Poly(ethylene glycol) Monoacrylate Immersed in Seawater[†]

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Biofouling of all structures immersed in seawater constitutes an important problem, and many strategies are currently being developed to tackle it. In this context, our previous work shows that poly(ethylene glycol) monoacrylate (PEGA) macromonomer grafted on preoxidized poly(methyl methacrylate) (PMMAox) films exhibits an excellent repellency against the bovine serum albumin used as a model protein. This study aims to evaluate the following: (1) the prevention of a marine extract material adsorption by the modified surfaces and (2) the antifouling property of the PEGA-g-PMMAox substrates when immersed in natural seawater during two seasons (season 1: end of April–beginning of May 2007, and season 2: end of October–beginning of November 2007). The antifouling performances of the PEGA-g-PMMAox films are investigated for different PEG chain lengths and macromonomer concentrations into the PEGA-based coatings. These two parameters are followed as a function of the immersion time, which evolves up to 14 days. The influence of the PEGA layer on marine compounds (proteins and phospholipids) adsorption is evidenced by time-of-flight secondary ion mass spectrometry (ToF-SIMS) and X-ray photoelectron spectroscopy (XPS). It was found that the antifouling efficiency of the PEGA-grafted surfaces increases with both PEGA concentration and PEG chain length.

Introduction

Marine fouling caused by organic and inorganic species is responsible for considerable damages to all devices and vessels immersed in seawater, and this induces a particular increase in maintenance costs.¹ Marine biofouling has been commonly prevented by using antifouling paints, which release toxic biocides such as cuprous oxide and organotin compounds.^{2–8} However, toxic ingredients contained in these paints were found to be ecologically very harmful, causing the feminization of different marine species in the world.⁹ Recently, several authors attempted to develop more environmentally friendly antifouling approaches to protect materials immersed in seawater.¹⁰ More precisely, different substances (e.g., terpenes, peptides, and sterols produced by marine organisms) have been already tested, and many of these compounds have shown to prevent fouling in filtered or simulated seawater.^{11–14} Devi et al. also reported that different extracts from marine invertebrates such as sponges and corals show a highly significant antifouling effect.¹⁵ Nevertheless,

implementation of the bioactive substances is still intricate, particularly because of high cost. Biofouling is the undesirable accumulation of microorganisms, plants, algae, and animals on submerged structures. The first step in biofouling is biofilm formation, which consists in adhesion of bacteria and diatoms. Bacteria and diatom adhesion is complex in nature and occurs on all artificial structures, including polymers.¹⁶ The main driving force for adhesion is the specific or nonspecific recognition of adsorbed organic material by bacteria.^{16–19} This material is defined as the conditioning film mainly composed of a protein layer (nitrogen compounds). On the other hand, bacteria adhering onto a surface usually secrete exopolymer and/or a protein matrix to cement themselves to the surface.²⁰ Currently, little is known about the initial forces evolved in the adhesion of bacteria to nonbiological surfaces. The adhesion of organisms is very complicated because a combination of modes is followed, not a single mode. The nature of bioadhesive interactions in the biofouling process is of main concern for many current research projects. Since the primary step in the biofilm formation mechanism involves protein or glycoprotein adhesives, several approaches are dedicated to produce protein-resistant surfaces^{21–23} able to avoid marine biofouling. Acrylamide-based

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polymers,²⁴ phospholipids,²⁵ and polysaccharides²⁶ are seen to relatively inhibit bioadhesion. However, the most promising polymer is poly(ethylene glycol) (PEG) or poly(ethylene oxide) (PEO).²⁷ Indeed, hydrophilic polymers such as PEG and PEO have shown both experimentally^{27,28} and theoretically^{29–31} to exclude adsorption of macromolecules. This is due to their high mobility, large exclusion volume, and steric hindrance effect of the highly hydrophilic layer. Our former work has shown that the competitive adsorption between a triblock PEO-PPO-PEO copolymer (Pluronic F68) and a protein (fibronectin) on polystyrene (PS) surfaces patterned by cold plasma oxidation allowed us to promote cell adhesion, proliferation, and growth only restricted to the PS treated surfaces where no Pluronic was present.^{32–35}

Concerning PEG grafting, several surfaces including polymers,^{36–39} glass,^{40,41} and stainless steel⁴² have been modified as models for protein or bacterial repellency. Kingshott et al.³⁹ grafted a PEG layer on poly(ethylene terephthalate) and on perfluorinated poly(ethylene-co-propylene) sheets by using radio frequency glow discharge. Indeed, the PEG-grafted substrates minimize the adsorption of a multicomponent protein solution containing lysozyme, human serum albumin, IgG, and lactoferrin.³⁹ The prevention of fibrinogen adsorption on polyethylene films was attempted by Holmberg et al.³⁸ First, they adsorbed ethylene oxide-propylene oxide block copolymer on pristine polyethylene (PE). In a second method, they grafted poly(ethylene glycol)-poly(ethylene imine) (PEG-PEI) copolymer to PE substrates. The better protein resistance of the PEG-PEI copolymer compared to that of the ethylene oxide-propylene oxide (EO-PO) block copolymer is attributed to a large entropy loss associated with protein adsorption on top of the loosely packed and highly mobile copolymer layer. Another contributing factor could be the fact that the EO-PO block copolymer, unlike the PEG-PEI graft copolymer, is not irreversibly bounded to the PE surface and may therefore be exchanged by fibrinogen. Some medical devices are elaborated by grafting an amphipathic ethylene glycol-butadiene block copolymers with different chain lengths to

dimethyldichlorosilane-coated glass.⁴⁰ The range of copolymer concentrations, which prevented platelet adhesion, was larger as the PEG chain length of the grafted copolymer became longer. Silanated polymers have been grafted onto various substrates such as silicates, silicon wafer, ceramic, aluminum,⁴³ stainless steel,^{44,45} and TiNi alloy.⁴⁶ As an example, glass surfaces modified with silanated PEGs reduce fibrinogen⁴¹ and γ -globulin (o) adsorption by more than 95% as compared with the control surfaces.

Gudipati et al.⁴⁷ reported that coatings based on amphiphilic cross-linked networks of hyperbranched fluoropolymers (HBFP) and poly(ethylene glycol), designated HBFP-PEG, with different PEG weight percentages inhibit protein (BSA, lectin from *Codium fragile*, lipopolysaccharides from *Escherichia coli*, and *Salmonella minnesota*) adsorption. Moreover, the amphiphilic cross-linked networks of hyperbranched HBFP-PEG prevent *Ulva* zoospore settlement as well as facilitate the zoospore release.

The marine antifouling and fouling-release performance of titanium surfaces coated with biopolymer inspired by marine mussels was investigated by Statz and workers.⁴⁸ The polymer is composed of methoxy-poly(ethylene glycol) (mPEG) conjugated to amino acid L-3,4-dihydroxyphenylalanine. The functionalized surfaces exhibit a substantial decrease in attachment of both cells of *Navicula perminuta* and zoospores of *Ulva linza* (from green algae) as well as the highest detachment of attached cells under flow compared to control surfaces.

In fact, despite the remarkable and proved efficiency of the PEG-grafted substrates^{49–55} toward several model proteins and/or isolated marine bacteria,^{56–58} the PEGylated surfaces have been tested only within protein solutions, and a limited number of specific bacteria growth was investigated. To the best of our knowledge, the test in natural seawater has not yet been reported.

In our previous study, poly(ethylene glycol) monoacrylate (PEGA) macromonomer was successfully photografted on preoxidized poly(methyl methacrylate) (PMMAox) films under UV irradiation.⁵⁹ The surface coverage of poly(ethylene glycol) was controlled by varying the PEGA concentration in the spin-coated solution. Inhibition of protein adsorption on the PEGA-g-PMMAox surfaces was studied by XPS and ToF-SIMS, using bovine serum albumin (BSA) as a model protein. The PEGA-

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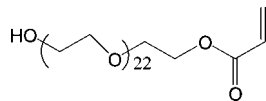


Figure 1. PEG monoacrylated (PEGA(1000)).

photografted PMMAox surfaces with high PEGA concentrations exhibit excellent protein repellency.

The aim of this study is the following: first, to test the PEGA-g-PMMAox surfaces toward a marine extract material (ME) as a model more complex than BSA, and second, to check the ability of these substrates to prevent biofouling in natural seawater. The influence on the fouling prevention of two parameters is reported. The PEGA concentration is varied into the coated formulations on PMMAox, so as the PEG chain length. ToF-SIMS is a technique already known for its highly sensitive analysis of molecules at polymer surfaces.^{60–63} It is used to detect adsorbed materials such as proteins and inorganic compounds after immersions in the ME solution and in natural seawater. The specific chemical features in the ToF-SIMS data related to the surface modifications of the PEGA-g-PMMAox are probed after immersions at two different periods in the year 2007 (seasons 1 and 2). After that, the spectral changes are elucidated using the principal component analysis (PCA) method. XPS is also used to determine from the nitrogen signal the amount of adsorbed species. The modifications of the surface morphology are illustrated by scanning electron microscopy (SEM).

Materials and Methods

Materials. Poly(methyl methacrylate) (PMMA) sheets with 0.5-mm thickness ($M_w = 198$ kDa) were purchased from Goodfellow. Before all surface modifications, pristine PMMA samples (1×1 or 1.5×1.5 cm² films) were washed with 2-propanol. The oxidizing procedure of PMMA films is described in ref 64.

Poly(ethylene glycol) monoacrylate with $M_n = 375$ (PEGA(375)), benzophenone (BZP), sodium chloride (NaCl), Na₂SO₄, deuterated trichloromethane, methanol, phosphorus pentoxide (P₂O₅), toluene, PEG-diol, diethyl ether, potassium oxide (KOH), dichloromethane (CH₂Cl₂), triethylamine (Et₃N), acryloyl chloride, acetone, and 2-propanol of analytical grade were obtained from Sigma Aldrich and were used as received. Poly(ethylene glycol) monoacrylate with average $M_n = 950$ –1050, noted PEGA(1000), was synthesized at UCL (see the PEGA(1000) chemical structure in Figure 1). Several formulations containing different amounts of PEGA(375) or PEGA(1000) macromonomers, BZP, and methanol were prepared. The BZP concentration was always kept at 0.7 wt % for all formulations. The solutions containing 10, 30, and 65 wt % PEGA(375) and a solution of 65 wt % PEGA(1000) were spin-coated on PMMAox (velocity of 5000 at 20 000 rpm/s, during 60 s). Then, the samples were UV-irradiated by a mercury lamp of medium pressure (80 W/cm) in ambient air. Finally, the samples were soaked in MilliQ water for four hours and dried under nitrogen flow.

The ME was purified and provided by the Université de Bretagne Sud as described below.

Marine Extract Material. The marine extract material was obtained from four strains of marine bacteria: *Pseudoalteromonas* sp. 3J6, *Bacillus* sp. 4J6, *Paracoccus* sp. 4M6,⁶⁵ and *Vibrio* sp.

D01.⁵⁸ Each strain was grown in 2 L of Marine Broth (Difco) during 48 h at 20 °C with shaking. The cells were recovered by centrifugation at 4000g and 4 °C for 10 min, washed in cold Tris HCl, 50 mM, pH 8.0, NaCl 100 mM, and resuspended in 40 mL of the same buffer containing one tablet of Complete Protease Inhibitor Cocktail Tablets (Roche Applied Science). The four cell suspensions were pooled, and bacteria were broken by three passages through a French press. Cell debris were removed by centrifugation at 14000g and 12 °C for 10 min. The supernatant was centrifuged again at 40000g and 4 °C for 1 h to pellet membranes and other insoluble materials. The supernatant was then incubated with RNase A (40 μg mL⁻¹) for 2 h at 37 °C. Complete RNA digestion was confirmed by agarose gel electrophoresis. The extract was dialyzed in dialysis membranes with a cutoff of 10 kDa against Tris HCl, 50 mM, pH 8.0, MgCl₂, 1 mM, and incubated during 20 min at 37 °C with 0.25 U mL⁻¹ of benzonase (Eurogentec) to digest DNA. The obtained product was then dialyzed against water and lyophilized to get the marine extract material.

The ME adsorption was performed using 2 mL of ME (0.2 mg/mL) solution at room temperature (19 °C) and pH = 7.39. The buffer used to prepare solution was phosphate-buffered saline (PBS). It was adjusted at pH = 7.39 and consisted in NaCl, 138 mM; KCl, 2.7 mM; KH₂PO₄, 6.5 mM; and Na₂HPO₄·2H₂O, 8.1 mM. The sample was deposited in a well of a tissue culture plate (Falcon 353225 from Becton Dickinson). After 6 h of adsorption, the sample was rinsed (four times 15 min) with MilliQ water. Afterward, each sample was flushed under nitrogen flow and stored in Petri dishes in a vacuum chamber (desiccator) at ambient temperature before analysis.

The PEGA-grafted PMMAox (PEGA-g-PMMAox) films were immersed for 6, 24, 72, and 336 h in circulating natural seawater at two different seasons in the year 2007 (season 1: end of April–beginning of May; season 2: end of October–beginning of November) at IFREMER-Brest.

Pristine PMMA and the PEGA(375)-g-PMMAox were immersed at both seasons, whereas the PEGA(1000)-g-PMMAox substrates were immersed only in season 2. The samples were aligned along a metallic wire through a hole situated at a corner of the film and immersed in a bath at IFREMER-Brest. The bath temperature was situated between 14 and 15 °C in April and 10 and 16 °C in November. The bath was filled with renewed seawater directly pumped from the Brest Bay. After immersion, the samples were rinsed (four times for 15 min) with sterile MilliQ water. Afterward, each sample was flushed under nitrogen flow and stored in Petri dishes in a vacuum oven at ambient temperature before analysis.

PEGA(1000) Synthesis. CH₂Cl₂ was distilled on P₂O₅. Et₃N was dried on KOH. PEG-diol, with $M_n = 950$ –1050, was purchased from Sigma-Aldrich and dried by azeotropic cycle with toluene.

Under argon, Et₃N (14 mL, 0.10 mol, 10 equiv) was added to a stirred solution of PEG-diol (10 g, 0.01 mol, 1 equiv) in CH₂Cl₂ (175 mL). Then, a solution of acryloyl chloride (0.812 mL, 0.01 mol, 1 equiv) in CH₂Cl₂ (25 mL) was added dropwise at 0 °C. The mixture was stirred during 48 h at 20 °C. The reaction medium was then washed with water. The organic layer was dried on Na₂SO₄, filtered, and concentrated in vacuum. The residue was precipitated in Et₂O, affording a white powder. The solid was dried for one night in vacuum (yield = 77.99% w/w).

NMR Analysis. ¹H NMR data were acquired on a Bruker Avance II 300 MHz. Chemical shifts are reported in parts per million relative to the residual solvent peak (CDCl₃). A dominant signal appeared at 96.91 ppm (figure not shown), which can be assigned to the glycol hydrogens of the ($n - 1$) repetitive units HO-(CH₂-CH₂-O) _{$n-1$} -CH₂-CH₂-O-C=O-CH=CH₂. A signal at 1.96 ppm is due to hydrogens of the first glycol unit linked to the ester group HO-(CH₂-CH₂-O) _{$n-1$} -CH₂-CH₂-O-C=O-CH=CH₂ and three weak signals at 1.000, 0.97, and 0.99 ppm, which are attributed to the hydrogens linked to the monomeric C=C double bond of the acrylate group HO-(CH₂-CH₂-O) _{$n-1$} -CH₂-CH₂-O-C=O-CH=CH₂ and a signal at 1 ppm of the H of HO-(CH₂-CH₂-O) _{$n-1$} -CH₂-CH₂-O-C=O-CH=CH₂. NMR results show clearly that PEGA was successfully obtained. The reaction yield is

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equal to 94% with respect to the PEG-diols. ^1H NMR: δ 6.45 (dd, 1H, $^2J = 1.4$ Hz, $^3J = 17.3$ Hz, $\text{CH}=\text{CH}_2$ trans), 6.15 (dd, 1H, $^2J = 10.4$ Hz, $^3J = 17.3$ Hz, $\text{CH}=\text{CH}_2$ trans), 5.83 (dd, 1H, $^2J = 1.4$ Hz, $^3J = 10.4$ Hz, $\text{CH}=\text{CH}_2$ cis), 4.31 (t, 2H, $^3J = 4.8$ Hz, $-\text{O}-\text{CH}_2-\text{CH}_2-\text{O}-\text{CO}-$), 3.64 (s, 92H, $-(\text{CH}_2-\text{CH}_2-\text{O})_n-$), 2.84 (t, 1H, $^3J = 5.8$ Hz, $-\text{CH}_2-\text{OH}$).

ToF-SIMS Analysis. The ToF-SIMS measurements were carried out with a PHI-Evans TFS-4000 MMI (TRIFT1) spectrometer using a pulsed 15 keV $^{69}\text{Ga}^+$ primary ion beam (800 pA DC, 8 kHz pulsing frequency, and 2-ns pulse width). The beam was rastered over a $120 \times 120 \mu\text{m}^2$ area. With a data acquisition time of 5 min, the fluence was 1.5×10^{12} ions cm^{-2} . The secondary ions were accelerated to 3 keV and focused by two lenses before undergoing 270° deflection in three hemispherical electrostatic analyzers. A 7 keV post acceleration was applied at the detector entry to increase the detection efficiency of high-mass ions. Charge effects were compensated by means of a 24 eV pulsed electron flood gun and a nonmagnetic stainless steel grid placed on each sample. Three spectra were recorded at different areas to check the reproducibility, and two sample sets were analyzed.

For each spectrum, the peak intensity was normalized to the total intensity (I_{total}) to eliminate differences in total secondary ion yield from spectrum to spectrum. The total intensity of a spectrum is the sum of all peak areas minus the ones of hydrogen and contaminant peaks, which are not very reproducible.

PCA was used to analyze the positive ion ToF-SIMS spectra. A description of the PCA principles applied to SIMS is given in ref 66. PCA was performed on all the peaks in the PMMA and PEGA-g-PMMAox spectra, before and after immersion in seawater, respectively. This treatment aims to discern the characteristic fragments of PMMA, PEGA, and the adsorbed molecules. Before PCA, the normalized data set is mean-centered. This operation centers the data set at the origin so that the variance in the data set is only due to differences in sample variance instead of differences in sample means.⁶⁷ PCA was performed with the Multion Software (Biophy Research).

Then, the biomolecule fragments and their relative intensity variations were analyzed to follow the changes in the surface composition due to the PEGA grafting extent onto PMMAox, depending on the experimental conditions.

X-ray Photoelectron Analysis. XPS spectra were recorded using a SSX-100 spectrometer (model 206 from Surface Science Instruments) equipped with an aluminum anode (10 kV, 20 mA) and a quartz monochromator. The charge stabilization was achieved using an electron flood gun at 6 eV and placing a grounded nickel grid 1 mm above the sample surface. The pass energy is 150 eV for survey analysis. Photoelectrons were collected at an angle of 55° with respect to the sample normal. All binding energies are referred to the neutral carbon C_{1s} peak at 284.8 eV.

Scanning Electron Microscopy. High-resolution images were obtained by field-effect gun scanning electron microscopy (982 Gemini from Leo). For morphological studies using SEM, the samples were covered by a thin layer of chromium (8.5 nm).

Results

This section is organized as follows. First, the results of surface characterization of the samples immersed in the ME solution are presented. Then, the effect of immersion in natural seawater on those sample surfaces during the two different seasons are shown and commented on.

The ME was first characterized by ToF-SIMS (Figure 2). The ToF-SIMS spectrum in the positive mode reveals the presence of inorganic fragments Na^+ ($m/z = 23$), Mg^+ ($m/z = 24$), and Ca^+ ($m/z = 40$) and organic ones, such as nitrogen containing peaks at $m/z = 70$, 72, and 86 and corresponding to $\text{C}_4\text{H}_8\text{N}^+$, $\text{C}_4\text{H}_{10}\text{N}^+$, and $\text{C}_5\text{H}_{12}\text{N}^+$, respectively. These peaks evidence the presence of neat amino acids and/or those induced by proteins

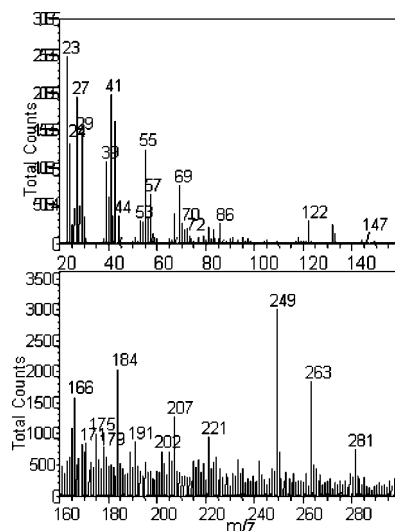


Figure 2. Positive ToF-SIMS data of marine extract material.

fragmentation in the ME. The presence of fragments from phosphatidylcholine (PDC) $\text{C}_5\text{H}_{13}\text{O}_3\text{NP}^+$ ($m/z = 166$) and $\text{C}_5\text{H}_{15}\text{O}_4\text{NP}^+$ ($m/z = 184$) indicates that compounds coming from the fragmentation of cell membranes constitute a non-negligible part of ME. In addition to a poly(dimethylsiloxane) (PDMS) contamination (peaks at $m/z = 73$, 147, 207, and 221), other peaks are observed at $m/z = 118$, 122, 132, 249, and 263, which are not yet identified. The negative spectrum (figure not shown) shows different peaks at $m/z = 26$, 42, 63, 79, 80, and 97 corresponding to CN^- , CNO^- , PO_2^- , PO_3^- , SO_3^- , and HSO_3^- fragments, respectively. These peaks reveal that the ME is not only composed from proteins but also contains phospholipids. The XPS analysis of ME gives the following atomic concentration ratio: C_{1s} (73.4%), O_{1s} (14.8%), N_{1s} (11.4%), and S_{2p} (0.5%), respectively (H is not detected). These XPS results agree well with the ToF-SIMS data.

After characterizing the marine extract material, we immersed the PEGA(375)-g-PMMAox films for 10, 30, and 65 wt % of PEGA(375) in an ME-based solution for 6 h as described in ref 59.

To follow the evolution of the ME-adsorbed amount on the different PEGA(375)-grafted PMMA-oxidized films, the ME positive normalized intensity (I_{ME^+}) is calculated by summing the peak intensities of the ME characteristic positive peaks (at $m/z = 70$, 72, 86, 166, and 184) and divided by the total positive intensity (I_{total^+}) in the spectrum. Figure 3 shows an important decrease in I_{ME^+} when PEGA(375) concentration rises from 0 to 30%. Above this value, I_{ME^+} decreases at a slower rate. A similar behavior is observed when the ME negative normalized intensity (I_{ME^-}) is calculated by summing the peak intensities of the ME negative peaks (at $m/z = 26$, 42, 63, 79, 80, and 97) and divided by the total negative intensity (I_{total^-}). These results constitute the first indication of the PEGA(375)-g-PMMAox surface ability to inhibit marine extract material adsorption. Since nitrogen is present in proteins and phospholipid molecules, the ToF-SIMS ME-normalized intensity can be correlated with the N/C ratio derived from XPS (Figure 3). More precisely, from 0 to 10% of PEGA(375), the N/C ratio decreases drastically. Above PEGA(375) 10% concentration, a slower decrease of the N/C ratio is observed. The difference in the decrease rate seen by XPS and SIMS could be explained by the different data treatment applied to the two methods.

(67) Wagner, M. S.; Pasche, S.; Castner, D. G.; Textor, M. *Anal. Chem.* **2004**, *76*, 1483.

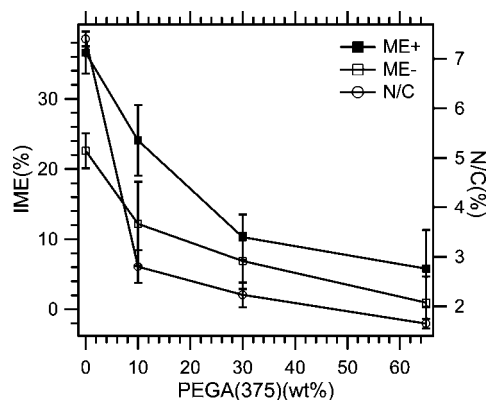


Figure 3. Variation of normalized I_{ME+} and I_{ME-} and N/C ratio, versus the PEGA(375) concentration (wt %).

For XPS, which is a quantitative method, the N/C ratio decrease is steeper because when N decreases due to the lower ME surface coverage, C increases due to the higher contribution of the polymer substrate.

For ToF-SIMS, the normalization is made on the total intensity in the spectrum, which is almost not influenced by the ME adsorption. The evolution of XPS nitrogen signal agrees well with the I_{ME+} and I_{ME-} (Figure 3) evolution as a function of PEGA(375) concentration. Both techniques show a protein and phospholipid resistance for PEGA(375)-g-PMMAox compared to the control substrates.

To study the effect of three days immersion in natural seawater at season 1, Figure 4 presents the positive SIMS spectra of PMMA and PEGA(375)-g-PMMAox, before and after immersion. Figure 4a,c compare the surfaces before immersion and Figure 4b,d after immersion, respectively.

For the pristine PMMA sample, the three-day immersion leads to a drastic decrease of the PMMA characteristic peak intensity at $m/z = 59$ and 69 (corresponding to $C_2H_3O_2^+$ and $C_4H_5O^+$, respectively) (Figure 4b), whereas new nitrogen-containing peaks appear at $m/z = 44, 56, 58, 70, 84,$ and 86 , corresponding to $C_2H_6N^+, C_3H_6N^+, C_3H_8N^+, C_4H_{10}N^+, C_5H_{10}N^+,$ and $C_5H_{12}N^+$, respectively. The nitrogenized peaks are similar to those already observed by Pradier et al.⁵⁸ when they investigated adhesion of three marine bacteria (DA, D01, and D41) on Teflon, stainless steel, or glass surfaces. Although the peak intensity seems to be weak, the amount of the adsorbed marine species is significant, as will be shown later by calculating their normalized positive intensity.

Consequently, the observed fragments in our study may mean that bacteria, and probably some microalgae (as will be illustrated by SEM images), are present on the substrates immersed in seawater. Hydrocarbon intense peaks (C_xH_y) are observed at $m/z = 55, 57, 91,$ and 97 . All the peaks cited above were also obtained on AISI 316 L stainless steel plates after being immersed in natural seawater at IFREMER, Brest Bay, in 2000.^{4,68}

Inorganic species such as Na^+ ($m/z = 23$), Mg^+ ($m/z = 24$), and Ca^+ ($m/z = 40$) are detected. Moreover, a PDC peak is detected at $m/z = 184$, attributed to the $C_5H_{15}O_4NP^+$ fragment and originating from cell membranes. There are also some fatty acid peaks ($m/z = 267, 282,$ and 341) corresponding to $C_{18}H_{35}O^+, C_{18}H_{36}ON^+,$ and $C_{21}H_{41}O_3^+$ fragments, respectively, and PDMS contamination peaks. Fatty acids (essentially from phytoplankton degradation) may account for approximately 4% of the dissolved organic carbon in natural waters.⁶⁹

After the pristine PMMA sample immersion, the negative spectra (figure not shown) confirm the presence of protein peaks such as CN^- ($m/z = 26$) and CNO^- ($m/z = 42$), while an important decrease of the characteristic PMMA peaks $C_3H_3O^-$ ($m/z = 55$), $C_4H_5O_2^-$ ($m/z = 85$), and $C_8H_{13}O_2^-$ ($m/z = 141$) occurs. Moreover, new peaks appear at $m/z = 58, 59.0143, 60,$ and 71 , which correspond to $C_2H_2O_2^-, C_2H_3O_2^-, CO_3^-,$ and $C_3H_3O_2^-,$ respectively. These peaks are characteristic of carbohydrates present on the PMMA surface immersed in seawater.

Peaks associated with inorganic compounds and their fragmentation are also detected: they are S^- ($m/z = 32$), SO^- ($m/z = 48$), SO_2^- ($m/z = 64$), sulfate SO_3^- ($m/z = 80$), HSO_3^- ($m/z = 81$), SO_4^- ($m/z = 96$), and HSO_4^- ($m/z = 97$). P^- ($m/z = 31$) and phosphate PO_2^- ($m/z = 63$) and PO_3^- ($m/z = 79$) fragments confirm the presence of phospholipids seen in the positive ToF-SIMS mode in addition to chloride Cl^- ($m/z = 35$ and 37) and bromide Br^- ($m/z = 79$ and 81) isotopes, respectively. Fatty acids are also seen in the negative mode at $m/z = 71$ ($C_3H_3O_2^-$), 227 ($C_{14}H_{27}O_2^-$), 255 ($C_{16}H_{31}O_2^-$), and 281 ($C_{18}H_{33}O_2^-$), respectively. Hydrocarbon peaks (C_xH^-) ($m/z = 15, 24, 25,$ and 38) and PDMS contaminations are still detected ($m/z = 59, 60, 73, 75, 149, 223,$ and 297).

Before immersion, the presence of grafted PEGA(375) on PMMAox is well evidenced in Figure 4c. Indeed, for 65 wt % of PEGA(375), the PEGA grafting leads to an intensity increase for many PEGA peaks such as $C_2H_3O_2^+$ ($m/z = 59$) and $C_4H_5O^+$ ($m/z = 69$) and several PEG peaks appear at $m/z = 45, 71, 73, 87,$ and 89 , which are attributed to $C_2H_5O^+, C_4H_7O^+, C_3H_5O_2^+, C_4H_7O_2^+,$ and $C_4H_9O_2^+,$ respectively. This information is better visualized by calculating the corresponding normalized positive intensities from PMMA ($I_{45} = 0.002, I_{59} = 0.012,$ and $I_{69} = 0.015$) and PEGA(375)-g-PMMAox ($I_{45} = 0.13, I_{59} = 0.022,$ and $I_{69} = 0.037$). The PEG peaks at $m/z = 45, 59, 69, 71, 73, 87,$ and 89 are more intense for 65 wt % PEGA(1000)-g-PMMAox surfaces than those for PEGA(375)-g-PMMAox films. It was not possible to measure the PEG thickness by ellipsometry because of the very close refractive index of PMMAox and PEGA. However, the PEGA average thickness estimated by AFM is about 44 ± 8 nm for PEGA(375) at high concentration (65 (wt %)). Thickness of the films formed by the longer PEGA(1000) macromonomer chains is expected to be higher than the one of PEGA(375) because polymerizable acrylic moieties are constant and the gyration radius is longer in case of PEGA(1000).

After immersion, the nitrogen-containing peak intensities decrease drastically (Figure 4d) and even the $C_5H_{15}O_4NP^+$ fragment is still completely missing.

To better visualize the effects of seawater immersion on the PMMA and PEGA(375)-g-PMMAox surfaces, PCA is realized on the positive ToF-SIMS data. PCA is performed on 222 peaks of the PMMA and PEGA(375)-g-PMMAox spectra before and after immersion. Figure 5a shows the principal component (PC1) score plot of positive SIMS spectra for PMMA and PEGA(375)-grafted PMMAox samples versus the immersion time. The PC1 scores capture 51.9% of the total variance in the data set and are influenced by the PEGA concentration in the grafted coatings. They allow us to discriminate the effect of the immersion time on the surface modifications. Indeed, for PMMA, PC1 is negative and it increases drastically to positive values with increasing immersion time from 0 to 72 h. Concerning PEGA(375)-g-PMMAox substrates, PC1 is less influenced by the immersion time and is still negative particularly for PEGA(375)-g-PMMAox with 65 wt % of PEGA(375), whatever the immersion period.

(68) Pradier, C. M.; Bertrand, P.; Bellon Fontaine, M. N.; Compère, C.; Costa, D.; Marcus, P.; Poleunis, C.; Rondot, B.; Walls, M. G. *Surf. Interface Anal.* **2000**, *30*, 45.

(69) (a) Norde, W. *Cells Mater.* **1995**, *5*, 97. (b) Thurman, E. M. *Organic Geochemistry of Natural Waters*; Kluwer Academic: Boston, 1985.

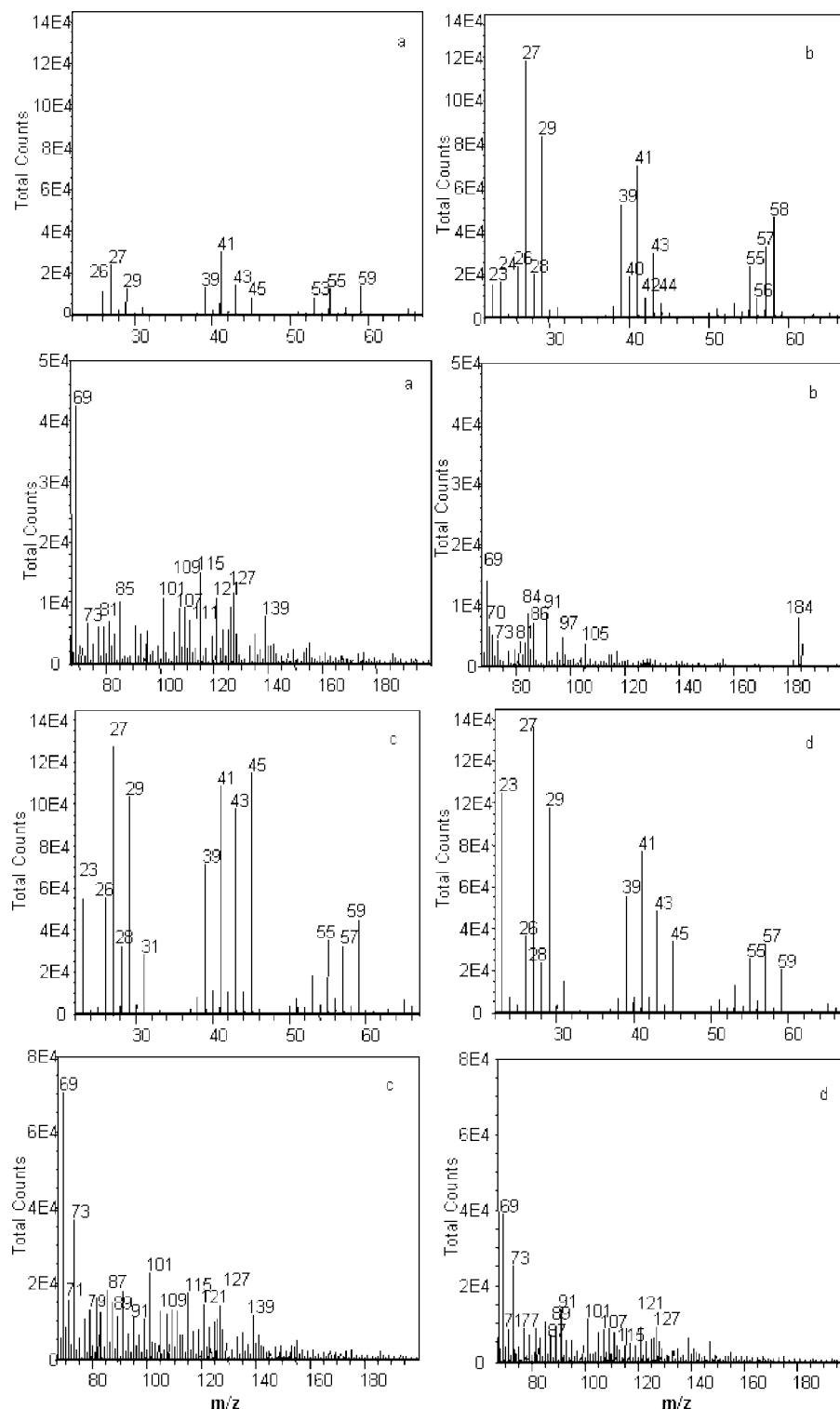


Figure 4. Positive ToF-SIMS data of PMMA (a) before (b) after immersion in seawater during 3 days and PMMAox grafted with PEGA(375) (65%) (c) before and (d) after immersion in seawater in season 1 for 3 days.

Looking at the PC1 loadings versus m/z (shown in Figure 5b) allows us to identify the main secondary ions that influence the sample discrimination seen in the (PC1) score plot. The PC1 loadings with negative values correspond to PEG characteristic fragments (at m/z = 45, 71, 73, 87, and 89), and the positive values correspond to protein and phospholipid fragments (at m/z = 44, 56, 58, 70, 84, 86, and 184), respectively. The PC2 score plot versus the time period (Figure 5c) explaining 14.4% of the total variance initially differentiates the PEGA-grafted PMMAox

from the nongrafted PMMA surfaces. The PC2 score increases with the PEGA(375) concentration in the coated formulation, showing the PEGA grafting extent increasing with the PEGA concentration. Finally, the PC2 loadings versus m/z are presented in Figure 5d. The PC2 loadings have to be considered in parallel with the PC2 scores versus the immersion time (Figure 5c). Indeed, the positive values of PC2 loadings correspond to PEG fragments, whereas the negative values are attributed to the PMMA fragments. The dispersion in the data may be explained

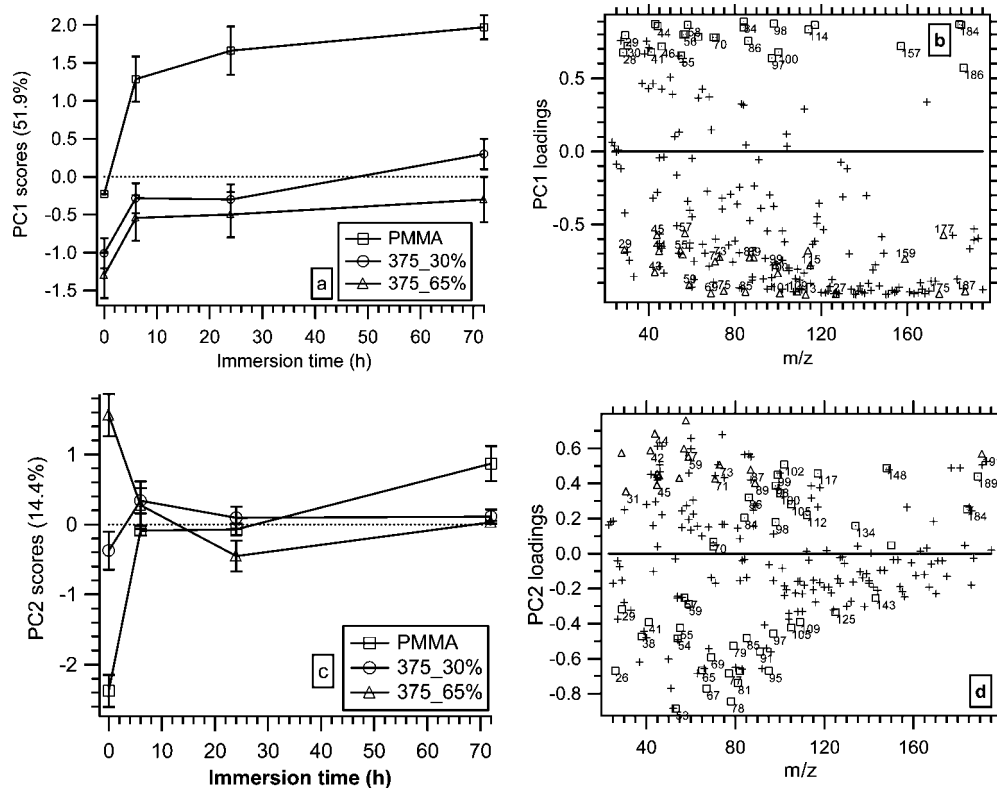


Figure 5. (a) PC1 scores plot from PCA of positive SIMS spectra of PMMA and PEGA(375)-grafted PMMAox samples versus immersion time. (b) PC1 loadings as a function of the m/z value of the detected secondary ions. (c) PC2 scores versus immersion time. (d) PC2 loadings as a function of the m/z values of the detected secondary ions.

by the fact that PEG chains are grafted onto preoxidized PMMA (PMMAox) and some fragments such as $m/z = 59$ and 69 are presented in both PMMA and PEG.

We see that PCA is sensitive to both the grafting of PEGA macromonomer and the adsorption of proteins and fragments from cell membranes. From Figure 5d, PEG characteristic fragments are observed at $m/z = 45, 59, 71, 73, 87$, and 89 , in addition to protein-related fragments at $m/z = 70, 84, 86, 98$, and 112 , and phospholipids ($m/z = 184$) are evidenced by the high positive PC2 values, whereas PMMA fragments at $m/z = 55, 57, 69, 85, 105, 125$, and 143 present negative PC2 values.

To follow the evolution of the adsorbed proteins (nitrogen compounds) and phospholipids, the normalized intensity of their characteristic fragments is evaluated. The positive normalized intensity (I_{pr+ph}) is obtained by dividing the sum of the corresponding fragment intensities (at $m/z = 44, 56, 58, 70, 84, 86$, and 184) by the total intensity (I_{total}) in the spectrum. Figure 6a presents a plot of (I_{pr+ph}) versus the immersion time in seawater in seasons 1 and 2. For pristine PMMA substrates immersed in both seasons, I_{pr+ph} increases drastically with the immersion time from 0 to 6 h. Above this immersion period, the I_{pr+ph} increases at a slower rate. When PEG chains are grafted to PMMAox, the I_{pr+ph} intensity is reduced. The increase of the PEGA(375) concentration from 30% to 65% leads to a stronger reduction of I_{pr+ph} .

Concerning the negative ToF-SIMS spectra, Figure 6b shows the normalized intensity (I_{CNO^-}) of the CNO^- protein characteristic fragment as a function of the immersion time for the two different PEGA concentrations. The I_{CNO^-} intensity increases with the immersion time for PMMA and PEGA-g-PMMAox substrates, whatever the used PEGA(375) concentration. However, the adsorption rate decreases when increasing the PEGA concentration. Nevertheless, for the first hours of immersion, low protein

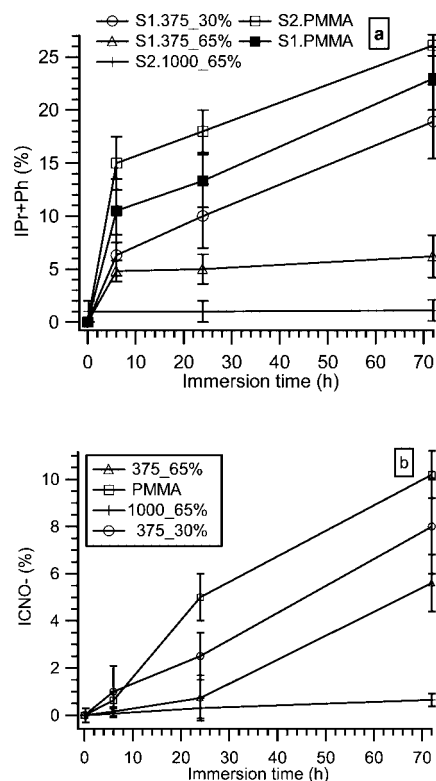


Figure 6. Variation of normalized (a) I_{pr+ph} and (b) I_{CNO^-} for PMMAox grafted with PEGA as a function of the immersion time at (S1) season 1 and (S2) season 2, respectively.

and phospholipid amount without saturation after 6 h is detected, especially on the PMMA substrates, and a slower adsorption rate is observed compared to that of the positive ToF-SIMS mode.

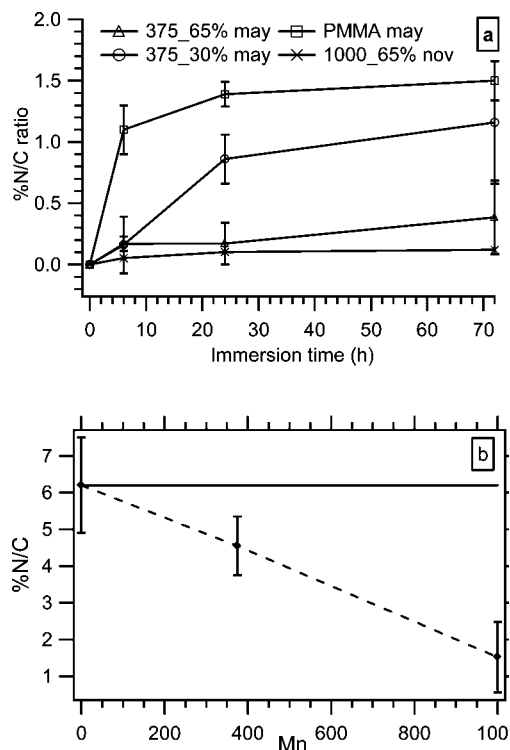


Figure 7. Evolution of the N/C ratio (a) of PMMA and PMMAox grafted with PEGA(375) at 30 and 65% versus the immersion time in season 1 and PMMAox grafted with PEGA(1000) at 65% in season 2. (b) Variation of the N/C ratio of PMMAox grafted with PEGA as a function of PEG weight average after immersion in seawater for 14 days in season 2.

This behavior may be explained by the fact that only one peak (CNO^-) is considered in the negative mode while several contributions (peaks) are taken into account in the positive mode to calculate the quantity of proteins and phospholipids adsorbed on the surface. The increase seen by ToF-SIMS for $I_{\text{pr+ph}}$ above the threshold of the XPS plateau could be related to the difference data treatment applied for both methods. In XPS, the N_{1s} signal would increase if the surface coverage increases with the immersion time. However, since the C_{1s} signal is also expected to change, this ratio could become constant.

To complement the ToF-SIMS results, the XPS N/C ratio is evaluated on the same samples and presented in Figure 7a. For pristine PMMA substrates, this N/C ratio increases rapidly with the immersion time from 0 to 6 h. Beyond this period, the nitrogen signal seems to saturate.

For PEGA(375), the N/C ratio is influenced by the grafting extent. It decreases for concentrations growing from 30 to 65%. For both concentrations, the N/C ratio also reaches a plateau, indicating a saturation in the protein and phospholipid adsorption kinetics. The onset of this plateau is directly related to the PEGA(375) concentration. It appears after six hours for the pristine PMMA control surfaces, whereas it is delayed up to 24 h for PMMAox grafted with formulation containing 30 wt % of PEGA(375). However, when PMMAox substrates are grafted with formulation containing 65 wt % of PEGA(1000) instead of PEG(375), the N/C ratio stays at zero whatever the immersion time at season 2 (Figure 7a). This result indicates clearly that longer PEG chains are more efficient to inhibit marine species adsorption than PEG(375).

All these SIMS and XPS data show that the PEG layer leads to a protein adsorption resistance.

To still improve the efficiency against marine fouling, PEGA(1000) instead of PEGA(375) was grafted on PMMAox

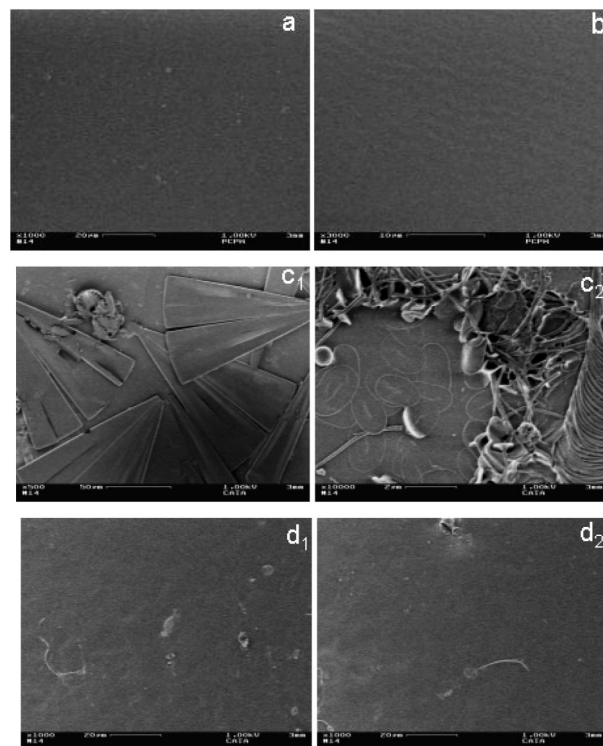


Figure 8. SEM images of (a) PMMA and (b) PEGA(1000)-g-PMMAox (65%) films before and (c₁,c₂) PMMA and (d₁,d₂) PEGA(1000)-g-PMMAox (65%) after 14 days of immersion in seawater in season 2.

(PEGA(1000)-g-PMMAox samples). Both sets were tested in seawater in season 2. The ToF-SIMS spectra of PEGA(375)-g-PMMAox are similar to those observed in season 1 for non-PEGylated surfaces immersed for 3 days in seawater. However, for PEGA(1000)-g-PMMAox films, neither protein nor PDC is detected even after 14 days of immersion. Only hydrocarbon contaminations ($m/z = 55, 57, 83, 97$, and 99) appear at the surface. Consequently, the $I_{\text{pr+ph}}$ intensity related to the 65% PEGA(1000)-g-PMMAox films, shown in Figure 6a, remains equal to zero whatever the immersion time from 0 to 72 h. Similarly in the negative mode, the CNO^- peak is not detected (Figure 6b) for the same immersion period.

XPS data were also obtained for these samples after the 14-day immersion period.

The dependence of the N/C ratio on the PEGA average molecular weight (M_n) after immersion in seawater for 14 days is shown in Figure 7b. In this figure, the reference ($M_n = 0$) corresponds to pristine PMMA. We see that the N/C ratio decreases from an initial average value of 0.062 for PMMA to 0.046 for 65% PEGA(375)-g-PMMAox, and down to 0.014 for the 65% PEGA(1000)-grafted coatings. As compared to that for pristine PMMA, an N/C loss of 77% ($\pm 2\%$) is obtained for PEGA(1000), greater than the one for PEGA(375), 26% ($\pm 3\%$).

Finally, Figure 8 displays SEM images of pristine PMMA and PEGA(1000)-g-PMMAox films before (Figure 8a,b) and after (Figure 8c,d) 14-day immersion in season 2. The images show that, after immersion and after washing, the pristine PMMA surface is partly covered by microorganisms such as diatoms species *Licmophora* sp. and *Surirella* sp. (Figure 8c₁) and *Prymnesiophyceae* scales (Figure 8c₂), whereas for the PEGA(1000)-grafted PMMAox film, only some aggregates are observed, probably due to the deposition of different salts present in seawater (Figure 8d).

Consequently, as seen by ToF-SIMS and XPS, longer PEG chains influence positively the prevention of protein and phospholipid adsorption and clearly inhibit adhesion of marine microorganisms such as bacteria and/or microalgae in natural seawater.

Discussion

This section is organized as follows. First, we present the results concerning the adsorption of the ME on pristine and PEGA-photografted preoxidized PMMA samples. Then, the behavior of those samples after immersion in natural circulating seawater is compared and commented on regarding the PEGA grafting density and PEGA chain length.

The positive effect of PEGA grafting to prevent protein and cell membrane fragment adsorption is clearly shown in Figure 3. We see a significant decrease of ME ToF-SIMS normalized intensity when increasing the grafting density. The XPS N/C ratio confirms this effect. The PEGA(375)-g-PMMAox substrates within 65% of PEGA(375) exhibit the lowest nitrogen content. This sample owns the higher ability to inhibit ME adhesion as compared to the coatings containing lower PEGA(375) concentrations (10 and 30%). These results are comparable to those previously obtained for BSA adsorption in the same experimental conditions.⁵⁹ They agree well with the known behavior of protein repellency markedly dependent on the hydrophilic/hydrophobic balance and on the mobility of the surface functional groups.

After we showed the relationship between the PEGylated PMMA surfaces and protein-suppression efficiency for a model protein (BSA)⁵⁹ and an ME, we tested the antifouling property of the PEGA coatings in natural seawater. A series of assays with different PEGA-grafted concentrations were performed at two different seasons. In such conditions, bacterial colonization has thus far been unavoidable.¹⁶

The complex ToF-SIMS results obtained on the immersed samples are discussed with the help of the data reduction statistical method, PCA.

PCA allows a differentiation of the two kinds of surfaces. As shown in Figure 5a, PC1 negative scores are associated with spectra of PEGA-g-PMMAox surfaces after immersion, whereas PC1 positive scores are related to spectra of pristine PMMA substrates after immersion. The examination of the PC1 loadings allows us to interpret these score differences in terms of chemical surface composition (Figure 5b). The fragments characteristic of marine species such as proteins and phospholipids have positive PC1 loadings, whereas the PEG characteristic fragments have negative PC1 loadings. The spectra with positive PC1 scores are strongly influenced by fragments with positive PC1 loadings, similar for the negative values. The combination of this score and loading information leads to the conclusion that the marine species are mainly adsorbed on the pristine PMMA surfaces.

The influence of the PEGA(375) grafting concentration on the PC1 scores for different immersion times is illustrated in Figure 9. For pristine PMMA, the PC1 score starts close to zero before immersion and increases to positive values with the immersion time. For PEGA(375) 30 wt % -g-PMMAox, the PC1 score starts from a negative value and grows to zero after a 6 h immersion, and it reaches a positive value after a 3-day immersion. However, for the 65% PEGA concentration, the PC1 score stays at negative values whatever the immersion period. As a consequence, the negative PC1 score values associated with the PEGA(375)-g-PMMAox can be considered as an indicator of its resistance toward marine species adsorption and/or adhesion. The PC2 scores are seen to be related to the PEGA grafting extent. Indeed, the PC2 scores increase with PEGA concentration.

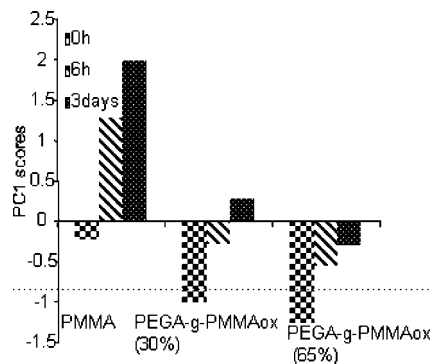


Figure 9. PC1 scores from PCA of positive SIMS spectra of PMMA and PEGA(375)-g-PMMAox samples versus the PEGA concentration (wt %) for different immersion times in season 1.

For the PEGylated surfaces, the very low ToF-SIMS-normalized intensities for proteins and phospholipids (Figure 6) evidence the protective effect provided by the PEGA grafting against the initial step of marine fouling.

The immersion of PEGA(375)-g-PMMAox samples in seawater shows the same tendency as that observed after either the model protein (BSA)⁵⁹ or the ME adsorption (Figures 3 and 6). The XPS N/C ratio is consistent with the evolution of the ToF-SIMS relative intensity (I_{pr+ph}) of the marine species as illustrated in Figure 6. The initial slope (from 0 to 6 h) of protein and phospholipid adsorption curves informs us about the affinity between the proteins and the surfaces.⁶⁹ This slope is steeper for the PMMA films than that for the PEGylated ones (Figures 6a and 7a), whatever the considered season and the PEG chain length. This means that the initial “colonization” rate is faster for the PMMA films than for the PEGA(375)-g-PMMAox substrates. The I_{pr+ph} curves show a continue and slow increase above 6 h of immersion only for PMMA and PEGA(375) 30 wt % -g-PMMAox, whereas the PEGA(375) 65 wt % -g-PMMAox samples exhibit a saturation.

The rapid colonization observed on pristine PMMA between 0 to 6 h immersion supports the work of Cunliffe et al.,²⁴ showing that the time period for protein adsorption and initial cell attachment is located between 1 and 24 h. The initial step of bacterial adhesion to the surface is the determinant for the biofilm formation.⁷⁰ We see that the PEGA(375)-g-PMMAox substrates exhibit a lower biomolecule adhesion. Indeed, after 6 h immersion, the XPS N/C ratio is about 10 times lower for PEGA(375)-g-PMMAox surfaces than that for PMMA. Moreover, this adhesion is still decreased by increasing the PEGA concentration.

These results on PEGA(375) layer resistance to protein and phospholipid adsorption agree well with recent studies showing the fouling resistance of coatings based on amphiphilic cross-linked networks of HBFP and poly(ethylene glycol). These coatings inhibit different protein adsorption and prevent the *Ulva* zoospore settlement as well as facilitate the zoospore release.⁴⁷ However, neither the pure HBFP coating (with the lowest surface energy) nor the PEG coating (known for its optimal protein resistance) nor their block copolymers allow the total fouling inhibition for proteins and lipopolysaccharides from *E. coli* and *S. minnesota*. The non-adhesion of the *Ulva* zoospores secreted by green alga *Ulva* prevents also surface colonization by the microalgae.

Not only the PEGA concentration but also the PEGA chain length is seen to play a role in the prevention of bioadhesion.

(70) Busscher, H. J.; Bos, R.; van der Mei, H. C. *FEMS Microbiol. Lett.* **1999**, *128*, 5229.

Indeed, we see that the PMMA surface grafting with longer PEG chains (1000 instead of 375) at the same concentration (65 wt %) allows a drastic decrease of the XPS N/C ratio and the biomolecule ToF-SIMS-normalized intensities after immersion in natural seawater in season 2 (Figures 6a and 7b). The low quantity of nitrogen species from PEGA(1000)-grafted surfaces compared to PEGA(375) ones confirms a significant decrease of marine fouling when PEG chain lengths are 2.6 times longer even after long immersion time. This shows the considerable improvement of fouling resistance when increasing the PEG graft density and chain length, as already claimed.⁷¹

Jeon et al.^{30,31} predicted by molecular studies the PEO or PEG chain length and graft density effects on protein resistance. Steric repulsion, van der Waals attraction, and hydrophobic interaction free energy are considered. In their model, the approach of a protein close to the PEG surface begins with a diffusion mechanism. The protein-PEG interactions depend on the van der Waals attraction between the PEG surface and protein through water. The further approach of the protein initiates a compression of the PEG chains, which induces a steric repulsion effect. The van der Waals attraction with the substrate decreases with increasing grafted PEG surface density and chain length. The steric repulsion free energy and the combined steric repulsion and hydrophobic interaction free energies are calculated as a function of the PEG surface density and chain length. This model predicts that a high surface density and long chain length of grafted PEO or PEG should exhibit optimal protein resistance. These predictions are well supported by our experimental results, showing that an excellent protection of the PEG-grafted surfaces is obtained at 65 wt % PEGA concentration with long PEG chain (PEGA(1000)) even for a long immersion period (14 days). Another work shows that PEG-electrografted glassy carbon and stainless steel surfaces at high PEG concentration reduce BSA and fibrinogen adsorption by more than 90%.⁵⁴

The PEG density depends on the spatial packing of the PEG chains.^{71,72} The packing mechanism is governed by the hydrodynamic (gyration) chain radius. This radius increases with the number of ethylene glycol units, reducing the space between neighboring chains until their radius of gyration overlaps. Consequently, longer chains allow a good coverage of the PEGylated surface and offer a better resistance against protein adsorption.

The best protein and phospholipid resistance reached for high PEG length ($M_n = 1000$) is quite probably due to remarkable PEG density realized in the present work. This suggests its high potential applicability as efficient nontoxic marine antifouling.

However, the reversible interactions between a bacterium and a substrate depend on the physical-chemical properties of the bacterial cells and the substrate surface as well as the experimental medium.⁵ It is found that not only the amount but also the conformation of proteins adsorbed on PEG layers is important

for cellular interactions.⁷³ Below 100-nm dry film thickness, marine organisms such as barnacles can cut through the underlying coats and thus establish a strong adhesion.

In the future, the PEGA(1000)-grafted films will be tested for a longer immersion period (several months). Higher PEG chain length with an average molecular weight of 5000, which is within the range typically used for protein resistance applications,^{27,30,74} will also be tested in natural seawater.

Conclusions

PEG with two different chain lengths was grafted on preoxidized PMMA films by photografting under UV irradiation. The antifouling property of the PEGylated surfaces was investigated first toward an ME, then by immersion in natural seawater at two different periods.

The inhibition of bioadhesion on the treated PMMA surfaces was evaluated by XPS and ToF-SIMS after different immersion times. These analyses reveal the presence of proteins and phospholipids on the control surfaces (pristine PMMA) after being immersed either in an ME solution or in natural seawater. They show that PEG-grafted films can reduce drastically the adsorption of ME and are able to prevent biofouling when immersed in natural seawater. This inhibition increases with PEG concentration.

The ToF-SIMS data treatment by PCA shows that the antifouling efficiency depends on both the PEG concentration and chain length. More precisely, the PCA analysis permitted us to identify and follow the surface modification occurring when the PEG acrylate concentration and the immersion time were changed. PC1 is a good indicator to quantify adsorbed marine compounds, and PC2 differentiates PEGA-g-PMMAox from nongrafted PMMA surfaces. The influence of PEG acrylate concentration on PC1 scores allows us to conclude that antifouling efficiency depends on PEG acrylate concentration. Moreover, marine species are mainly adsorbed on native PMMA. From the normalized ToF-SIMS intensity of proteins and phospholipids, it appears clear that the antifouling efficiency of the PEGylated substrates depends on both PEG acrylate concentration and PEG chain length.

The best results were obtained for PEG chain length of $M_n = 1000$ and 65 wt % in solution, where biomolecule signals were not observed, indicating the absence of diatoms and/or microalgae compared to PMMA even after a 14-day immersion period.

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